Supplementary Notes: Teacher-Student Distillation

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*Note. The material in these supplementary notes was covered in the 16-week version of this course and is **optional reading** for the 8-week offering. It assumes the Week 6 lecture notes on transformers (self-attention, positional encoding, the transformer block, and next-token training) and is provided for students who want to see how those ideas are extended in current research systems.*

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1 Conceptual Summary

Reference: G. Hinton, O. Vinyals, J. Dean, "Distilling the Knowledge in a Neural Network," 2015, arXiv:1503.02531.

- **Goal:** Transfer a distribution $q_\phi(\cdot \mid c)$ learned by a large/accurate *teacher* to a smaller/faster *student* $p_\theta(\cdot \mid c)$.
- **Core idea:** Make the student match the teacher's *soft targets* (probability distributions, geometry distributions, or denoising trajectories), optionally combined with ground-truth *hard targets* and/or *intermediate feature* matching.
- **Why it works:** Soft targets encode "dark knowledge" (relative similarities among outcomes) that hard labels don't convey; this often improves generalization and compresses large models effectively.

2 Unified Objective

Let

- c = context (input x, history $y_{<t}$, conditioning variables),
- o = predicted object (class, token, structure, noise, score, ...),
- $q_\phi(o \mid c; T)$ = teacher distribution with temperature $T > 1$,
- $p_\theta(o \mid c; T)$ = student distribution.

We minimize:

$$\min_\theta \ \mathbb{E}_{c\sim\mathcal{D}}\Big[\alpha\, \ell_{\text{hard}}(c;\theta) + \beta\, T^2\, D\big(q_\phi(\cdot \mid c;T) \,\|\, p_\theta(\cdot \mid c;T)\big) + \gamma\, R_{\text{feat}}(c;\theta,\phi) + \delta\, R_{\text{traj}}(c;\theta,\phi)\Big].$$

Components:

- ℓ_{hard}: ground-truth loss (CE/NLL/MSE). Set $\alpha = 0$ for pure distillation.
- D: divergence (KL, reverse-KL, JS, Wasserstein, energy distance).
- R_{feat}: hidden state matching, e.g. $\|h_T(c) - h_S(c)\|^2$.
- R_{traj}: trajectory matching (diffusion timesteps, autoregressive rollouts).

3 Canonical Instantiations

3.1 Classifiers

$$\mathcal{L} = \alpha\, \text{CE}(y, p_\theta(\cdot \mid c;1)) + \beta\, T^2\, \text{KL}\big(q_\phi(\cdot \mid c;T) \,\|\, p_\theta(\cdot \mid c;T)\big).$$

3.2 LLMs (token-level)

$$\mathcal{L} = \sum_t \Big[\alpha\,(-\log p_\theta(y_t \mid c)) + \beta\, T^2\, \text{KL}\big(q_\phi(\cdot \mid c;T) \,\|\, p_\theta(\cdot \mid c;T)\big)\Big].$$

3.3 Protein Structure Prediction

$\mathcal{L} = \beta\, D(q_\phi(R \mid \text{seq}), p_\theta(R \mid \text{seq})) + \gamma\, \|h_T - h_S\|^2 + \alpha\, \ell_{\text{hard}}$ (e.g., RMSD/MSE to ground truth).

3.4 Diffusion / Score-Based Models

$$\mathcal{L} = \mathbb{E}_{t,x_t}\Big[\|\epsilon_T(x_t,t,c) - \epsilon_\theta(x_t,t,c)\|^2\Big] \quad \text{or} \quad \mathbb{E}\Big[\|s_T - s_\theta\|^2\Big].$$

4 Proofs and Derivations

4.1 KL and Cross-Entropy Equivalence

$$\mathrm{KL}(q\|p) = \sum_i q_i \log \frac{q_i}{p_i} = H(q,p) - H(q).$$

Since $H(q)$ is constant in p, minimizing KL is equivalent to minimizing cross-entropy.

4.2 Temperature Scaling and T^2 Factor

Let $p = \mathrm{softmax}(z/T)$.

$$\frac{\partial \tilde{\mathcal{L}}}{\partial z_j} = \frac{1}{T}(p_j - q_j).$$

Multiplying by T^2 rescales gradients:

$$\frac{\partial (T^2 \tilde{\mathcal{L}})}{\partial z_j} = T(p_j - q_j).$$

4.3 Sequence-Level KL Decomposition (LLMs)

$$\mathrm{KL}\big(q(y_{1:T} \mid x) \,\|\, p(y_{1:T} \mid x)\big) = \sum_{t=1}^{T} \mathbb{E}_{q(y_{<t}|x)}\Big[\mathrm{KL}\big(q(y_t \mid y_{<t}, x) \,\|\, p(y_t \mid y_{<t}, x)\big)\Big].$$

4.4 MSE Minimizer is Conditional Expectation

For o, c:

$$g^*(c) = \arg\min_g \mathbb{E}[\|o - g(c)\|^2] = \mathbb{E}[o \mid c].$$

4.5 Gradient View of Distillation

$$\nabla_\theta \mathcal{L}_{\text{distill}} = -\mathbb{E}_c \mathbb{E}_{o\sim q(\cdot|c)}\big[\nabla_\theta \log p_\theta(o \mid c)\big].$$

This is maximum likelihood under teacher samples.

5 Practical Knobs

- Balance weights $\alpha, \beta, \gamma, \delta$.
- Divergence choice: $\mathrm{KL}(q\|p)$ (mode covering) vs. $\mathrm{KL}(p\|q)$ (mode seeking).
- Temperature T: higher T softens targets; T^2 scaling stabilizes gradients.
- Context choice c: teacher-forcing vs. student-rollout, diffusion schedules, etc.

6 Pseudocode Sketches

6.1 LLM Token-Wise Distillation

```
for (x, y) in data:
    with no_grad():
        q_logits = teacher(x, y_inp=y[:-1])
        q = softmax(q_logits / T)
    p_logits = student(x, y_inp=y[:-1])
    p = softmax(p_logits / T)

    loss_hard = CE_hard(y, logits=student(x, y_inp=y[:-1]))
    loss_soft = KL(q, p) * (T**2)
    loss = alpha*loss_hard + beta*loss_soft
    loss.backward(); opt.step()
```

6.2 Diffusion

```
for x0 in data:
    t ~ Uniform({1,...,T_steps})
    eps ~ Normal(0, I)
    x_t = forward_diffuse(x0, eps, t)

    eps_T = teacher.predict_noise(x_t, t)
    eps_S = student.predict_noise(x_t, t)

    loss = mse(eps_S, eps_T)
    loss.backward(); opt.step()
```

6.3 Protein Structures

```
for seq, gt in data:
    with no_grad():
        q_bins, h_T = teacher(seq)
    p_bins, h_S = student(seq)

    loss_kl  = KL(q_bins, p_bins)
    loss_feat = mse(h_S, h_T)
    loss_gt   = rmsd(student_coords(seq), gt_coords)
    loss = beta*loss_kl + gamma*loss_feat + alpha*loss_gt
    loss.backward(); opt.step()
```

One-Line Contrast with GANs

Teacher-student minimizes a fixed divergence to a teacher (cooperative), while GANs define an adversarial game with a discriminator-driven divergence.

7 Teacher-Student with RLHF (Fixed Reward Model)

Reinforcement Learning with Human Feedback (RLHF) can be expressed in the teacher-student framework when the reward model is fixed.

7.1 Setup

- **Student (policy):** $\pi_\theta(y \mid x)$.
- **Teacher (reward-shaped distribution):**

$$\pi^*(y \mid x) \propto \pi_{\text{ref}}(y \mid x)\, \exp\Big(\tfrac{1}{\beta} r_\phi(x, y)\Big),$$

  where π_{ref} is a reference policy and r_ϕ is the fixed reward model.
- **Goal:** Distill π^* into π_θ by minimizing KL divergence.

7.2 Objective

We maximize the reward-regularized objective:

$$\max_\theta \; \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi_\theta(\cdot|x)}\Big[r_\phi(x, y) - \beta \log \tfrac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}\Big].$$

Equivalently, the optimal distribution is

$$\pi^*(y \mid x) \propto \pi_{\text{ref}}(y \mid x)\, \exp\Big(\tfrac{1}{\beta} r_\phi(x, y)\Big).$$

The student then minimizes:

$$\mathcal{L}_{\text{RLHF}}(\theta) = \text{KL}\big(\pi^*(\cdot \mid x) \,\|\, \pi_\theta(\cdot \mid x)\big).$$

7.3 Gradient Derivation

Expanding the gradient:

$$\nabla_\theta \mathcal{L}_{\text{RLHF}} = -\,\mathbb{E}_{y\sim\pi^*(\cdot|x)}\big[\nabla_\theta \log \pi_\theta(y \mid x)\big].$$

Since π^* may be intractable to sample exactly, we use a policy-gradient estimator:

$$\nabla_\theta J(\theta) = \mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\Big[\big(r_\phi(x, y) - \beta \log \tfrac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}\big)\, \nabla_\theta \log \pi_\theta(y \mid x)\Big].$$

7.4 Interpretation

- The **teacher** is the reward-shaped distribution π^*.
- The **student** is π_θ, trained to approximate π^*.
- The **KL term** acts as a distillation regularizer, keeping π_θ close to π_{ref}.
- This shows RLHF with a fixed reward model is a form of *teacher-student distillation*, where human preference (via r_ϕ) modifies the teacher distribution.

7.5 Pseudocode (Fixed-Reward RLHF)

```
# Inputs:
#   D            : supervised prompts/distribution over x
#   pi_ref       : frozen reference policy
#   r_phi        : fixed reward model (trained from human preferences)
#   pi_theta     : trainable policy (student)
#   beta         : KL regularization strength
#   eta_ent      : (optional) entropy bonus coefficient
#   b(x)         : (optional) baseline/critic to reduce variance

for epoch in 1..E:
    for minibatch x ~ D:
        # 1) Sample responses from current policy
        y ~ pi_theta(. | x)                 # e.g., nucleus/temperature sampling

        # 2) Compute rewards and logs
        r = r_phi(x, y)                     # scalar reward
        logp_theta = log pi_theta(y | x)
        logp_ref   = log pi_ref(y | x)

        # 3) Form KL-regularized advantage (reward shaping)
        #    A = r - beta * (log pi_theta - log pi_ref) - b(x)
        #    (b(x) can be moving average or learned value baseline)
        A = r - beta * (logp_theta - logp_ref) - b(x)

        # 4) Policy loss (REINFORCE-style), optional entropy bonus
        loss_policy = - mean( A * logp_theta )
        loss_entropy = - eta_ent * mean( entropy(pi_theta(. | x)) )
        loss = loss_policy + loss_entropy

        # 5) Optimize student policy
        loss.backward()
        optimizer.step(); optimizer.zero_grad()

        # (Optional) Monitor forward KL to reference:
        #   KL_forward = mean( KL(pi_theta(.|x) || pi_ref(.|x)) )
        # (Optional) Use PPO-style clipping on ratio exp(logp_theta - logp_old)
```

Notes.

- The KL penalty appears inside the shaped advantage $A = r_\phi - \beta(\log \pi_\theta - \log \pi_{\text{ref}})$, which yields the same stationary point as minimizing $\mathrm{KL}(\pi^* \| \pi_\theta)$.
- A baseline $b(x)$ reduces gradient variance (e.g., moving average or learned value head).
- Entropy bonus encourages exploration and prevents early collapse.
- PPO clipping, adaptive β, or trust-region updates can further stabilize training.